

Business Requirements Document (BRD)

Employee Leave Management & Approval Portal (ELAP)

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Reviewed By	Sameer Rathod (Project Manager)
Approved By	Prof. Mr. Sayantan (Project Sponsor)
Date	August 29, 2026
Institution	ITM Skills University — Project Management Course
Client	Prof. Mr. Sayantan (acting as Business Sponsor / HR Leadership)

Team

Name	Role
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Sameer Rathod	Project Manager
Sanchita Suryawanshi	Scrum Master
Soham Ahirrao	Business Analyst
Antriksh Manwadkar	Solution Architect
Atharva Gahine	Tech Lead

Document Revision History

Version	Date	Author	Change Description
1.0	Aug 29, 2026	Soham Ahirrao (BA)	Initial BRD — complete requirements, tech stack, architecture, timeline, and all decisions finalized

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1. Executive Summary

The client organization currently manages all employee leave requests, approvals, and tracking through a combination of email threads and Microsoft Excel spreadsheets. This fragmented approach results in lost requests, approval delays, payroll discrepancies, and a complete lack of real-time visibility for the HR department across 8 departments.

This document defines the business requirements for the **Employee Leave Management & Approval Portal (ELAP)** — a centralized web application that will replace the existing manual process with a structured digital workflow. The portal will enable employees to submit leave requests online, route those requests through a **2-tier approval chain** (Reporting Manager → HR), and provide HR with a real-time dashboard covering all departments.

The system will be built as a modern **Next.js** full-stack web application with a **PostgreSQL** database, deployed on **Vercel** (application) and **Supabase/Neon** (database), using **Resend** for email notifications.

The project is scoped exclusively to **leave management** for Phase 1. Adjacent HR functions (payroll, recruitment, performance management) are explicitly out of scope.

[!IMPORTANT] Compressed Timeline: This project must be delivered in **5 weeks** (August 29 – September 30, 2026) using **1-week sprints**. The scope and priorities have been calibrated accordingly — P0 features are non-negotiable, P1 features are included only if time permits.

Project Budget: ₹1,00,000 (One Lakh INR — hypothetical) **Actual Estimated Cost:** ₹500 – ₹3,000 (free-tier stack) **Timeline:** 5 weeks | 5 sprints | August 29 – September 30, 2026

2. Business Problem Statement

2.1 Problem Description

The organization manages employee leave through **email requests and Excel spreadsheets**. There is no centralized system for leave application, approval routing, or organizational visibility.

#	Problem	Impact
1	No single source of truth — leave data fragmented across email inboxes and multiple Excel files	Conflicting records, data loss, no reliable history
2	Approval bottlenecks — requests buried in manager email inboxes	Average approval time: 3–5 business days
3	HR blindness — no real-time visibility across 8 departments	Cannot monitor trends, pending approvals, or compliance
4	No audit trail — no record of who approved what and when	Compliance risk, dispute resolution impossible

5	Manual payroll reconciliation — HR manually cross-references leave data for payroll	Errors in 5–10% of salary calculations monthly
6	No leave balance tracking — employees and managers have no visibility into remaining entitlements	Over-utilization, disputes, manual counting
7	Scattered communication — leave discussions across disconnected email threads	Context lost, repeated back-and-forth

2.2 Cost of Inaction (Indian Industry Benchmarks)

Impact Area	Estimated Cost / Waste
HR hours wasted on manual Excel tracking	8–12 hours per week
Manager time on email-based reviews and disputes	3–5 hrs/week per dept (×8 depts = 24–40 hrs/week org-wide)
Payroll errors from manual reconciliation	5–10% of salary calculations affected monthly
Annual hidden cost for a mid-size Indian company	₹12–21 Lakhs/year
Employee replacement cost (attrition from frustration)	6–9 months of salary per departing employee

2.3 Affected Stakeholders

Stakeholder	Impact
Employees (all 8 departments)	No centralized submission, no status tracking
Reporting Managers (8 departments)	Requests lost in email, no team view, manual follow-ups

HR Department	Zero real-time visibility, manual reconciliation, compliance risk
Finance / Payroll	Inaccurate leave data → incorrect salary processing
Leadership	No data-driven workforce planning

3. Business Objectives (SMART)

#	Objective	Measurable Target	Timeframe
OBJ-01	Eliminate Excel-based leave tracking	100% of leave requests via portal; 0 active Excel trackers	Within 2 weeks of go-live
OBJ-02	Reduce leave processing time	From 3–5 business days → ≤ 48 hours	Operational from Sprint 3
OBJ-03	Achieve complete HR visibility	All 8 departments on HR dashboard with real-time status	Delivered by Sprint 3
OBJ-04	Establish 100% auditable leave trail	Every status change logged with timestamp, actor, and action	Built into every module
OBJ-05	Reduce HR reconciliation effort by 70%	Centralized data eliminates manual cross-referencing	Measurable within 30 days of go-live
OBJ-06	Zero learning curve for employees	First leave request completable without training in ≤ 3 clicks	UAT validation

4. Success Metrics & KPIs

KPI	Target	How Measured
Portal adoption	100% within 2 weeks of go-live	Usage logs vs. employee count
Average processing time (submission → final approval)	≤ 48 hours	System timestamps
Excel tracker elimination	0 active trackers	HR audit
Data accuracy	≥ 99%	Database audit
HR dashboard coverage	100% of departments, real-time	Functional testing
Manager first-response time	≤ 24 hours	System timestamps
System uptime	≥ 99% during business hours	Vercel monitoring
Ease of use	≤ 3 clicks to submit	UX testing

5. Project Scope

5.1 In Scope — Phase 1

[!IMPORTANT] Timeline reality check: With only 5 weeks, **P0 features are the hard commitment**. P1 features will be delivered on a best-effort basis. If the team falls behind at any sprint review, P1 features will be the first to be cut — as agreed with the Project Manager (Sameer Rathod).

#	Feature / Module	Priority	Sprint
1	Employee authentication (email/password) with RBAC (Employee, Manager, HR)	P0	Sprint 1

2	Employee profile (Employee ID, name, department, reporting manager, joining date)	P0	Sprint 1
3	Leave request creation (leave type, dates, reason)	P0	Sprint 2
4	Leave type and balance tracking (CL: 12, SL: 12, EL: 15 days/year)	P0	Sprint 2
5	2-tier approval: Reporting Manager → HR final approval	P0	Sprint 2
6	Manager dashboard (pending requests, approve/reject with comments)	P0	Sprint 2
7	HR dashboard (org-wide visibility, dept/status/date filters)	P0	Sprint 3
8	Leave status tracking for employees	P0	Sprint 2
9	Leave history and records	P0	Sprint 3
10	8 departments — dynamically configurable	P0	Sprint 1
11	Audit trail (all status transitions logged)	P0	All sprints
12	Leave request cancellation (before HR final approval)	P0	Sprint 2
13	Responsive web design	P0	All sprints
14	Automated email notification on submission + decisions	P1	Sprint 3
15	Email with Accept/Reject action links (signed JWT)	P1	Sprint 3
16	Rejection reason/comment by manager/HR	P1	Sprint 2
17	Attachment upload (PDF/JPG/PNG, max 5 MB)	P1	Sprint 3
18	In-portal comment thread per leave request	P1	Sprint 3
19	Search, filter, sort across all views	P1	Sprint 3

20	Department-wise reports (CSV export)	P1	Sprint 3
21	Manager auto-reminder (48h) and HR escalation (72h)	P1	Sprint 3

5.2 Out of Scope — Phase 1

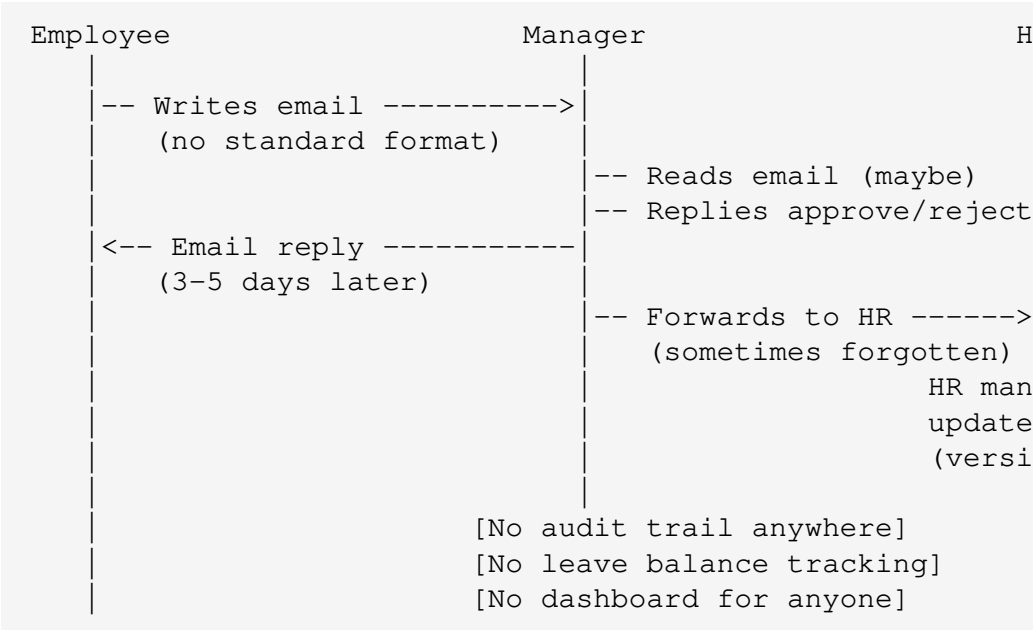
#	Feature	Reason
1	Payroll / salary processing	Client confirmed: leave-only
2	Recruitment, onboarding, performance management	Not leave management
3	Attendance / time tracking	Phase 2
4	Training management	Not leave management
5	Advanced analytics / predictive dashboards	Phase 2
6	Native mobile app (iOS/Android)	Web app is responsive; native app = Phase 2
7	Full HRMS suite	Client confirmed: additive, not replacement
8	Third-party integrations (SAP, PeopleHR)	Not required
9	SSO / Active Directory	Standalone auth sufficient

6. Stakeholder Register

Stakeholder	Name	Role	Interest	Influence
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Project Sponsor / Client	Prof. Mr. Sayantan	Business Sponsor, final sign-off authority	High	High
Project Manager	Sameer Rathod	Scope, timeline, delivery, stakeholder coordination	High	High
Scrum Master	Sanchita Suryawanshi	Sprint process, ceremonies, blocker removal	High	Medium
Business Analyst	Soham Ahirrao	Requirements, user stories, acceptance criteria, UAT	High	High
Solution Architect	Antriksh Manwadkar	System architecture, tech decisions, security	High	High
Tech Lead	Atharva Gahine	Implementation, code quality, deployment	High	High
Employees (simulated)	—	End users submitting leave	High	Low
Reporting Managers (simulated)	—	Tier 1 approvers	High	Medium
HR (simulated)	—	System owner, Tier 2 approver, org-wide monitoring	High	High

7. Current State Process (As-Is)

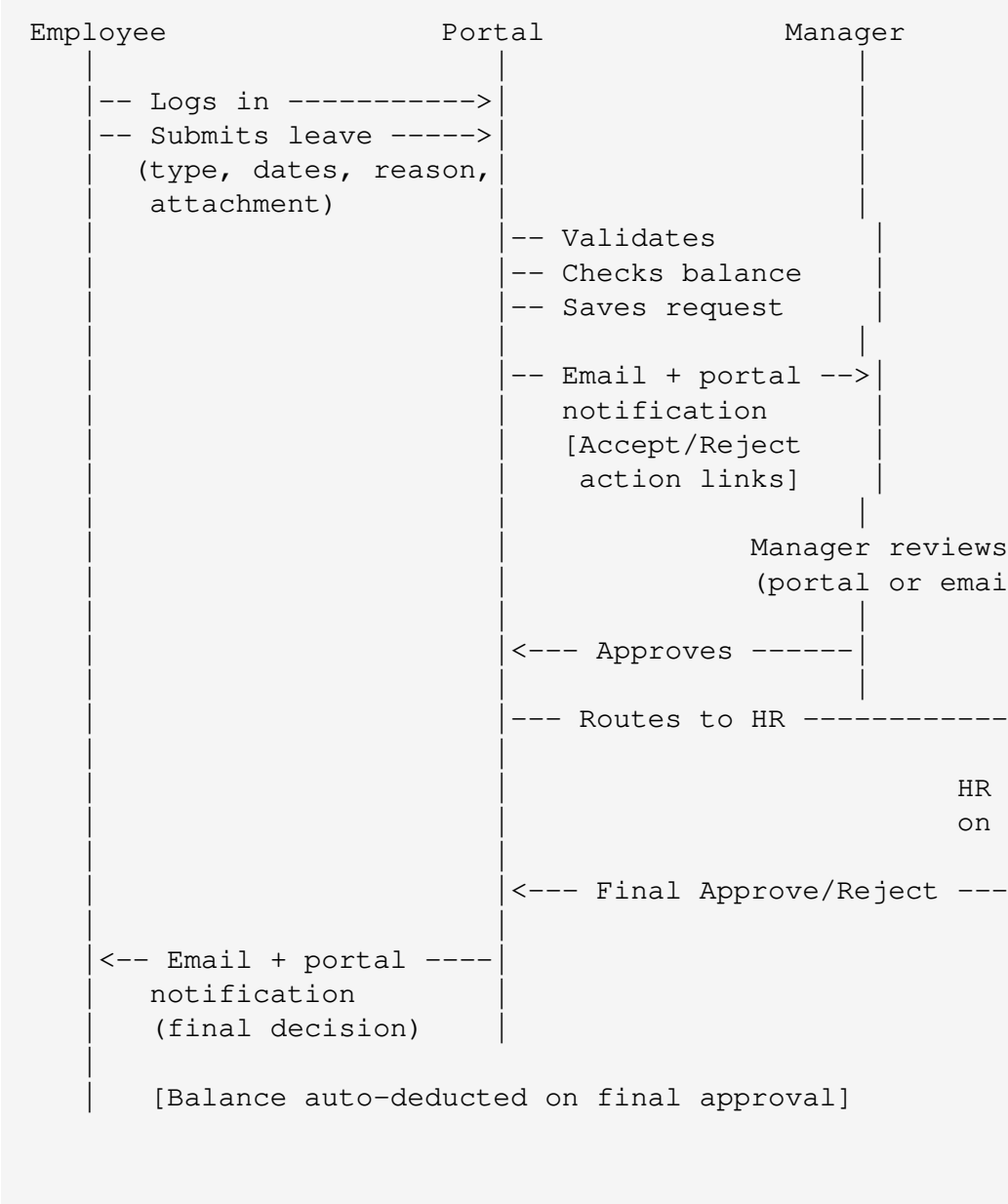


Pain Points

#	Pain Point	Severity
1	No standard format for leave requests	High
2	Requests buried in manager's inbox	High
3	Each department maintains its own Excel	Critical
4	HR has zero real-time visibility	Critical
5	No audit trail	High
6	8–12 hrs/week wasted on reconciliation	High
7	No leave balance visibility	Medium
8	Discussions scattered across email threads	Medium

8. Future State Process (To-Be)

2-Tier Approval Flow



[Audit trail at EVERY step]

[HR Dashboard shows everything, all 8 depts]

Escalation Path (Automated — P1)

Trigger	System Action
Manager no action within 48 hours	Auto-reminder email to Manager
Manager no action within 72 hours	Request flagged as "Escalated" on HR Dashboard
HR no action within 48 hours after manager approval	Auto-reminder email to HR

9. User Personas & Needs

9.1 Employee

Attribute	Detail
Who	Any employee across 8 departments
Goal	Submit leave in ≤ 3 clicks, track status without follow-ups
Pain Today	Emails, waiting, no balance visibility, no status tracking
Needs	Simple form, balance display, real-time status, history, notifications

9.2 Reporting Manager

Attribute	Detail
Who	Department head or team lead
Goal	Quickly approve/reject from dashboard or email
Pain Today	Requests buried in inbox, no team view
Needs	Pending-request dashboard, one-click approve/reject, rejection comments

9.3 HR Administrator

Attribute	Detail
Who	HR team member responsible for org-wide leave management
Goal	Monitor all activity, give final approval, maintain accurate records
Pain Today	8+ Excel files, manual reconciliation, no audit trail
Needs	Org-wide dashboard, filters, final approval, exportable reports, audit trail

10. Functional Requirements

Req ID	Requirement	Description	Priority
FR-01	Secure Authentication	Email/password login. Three roles: Employee, Manager, HR.	P0

FR-02	Role-Based Access Control	Employees see own data. Managers see their team. HR sees all.	P0
FR-03	Employee Profile	Employee ID, name, email, department, reporting manager, joining date.	P0
FR-04	Leave Request Creation	Select leave type (CL/SL/EL), start date, end date, reason.	P0
FR-05	Leave Balance Display	Show current balance by type before and after submission.	P0
FR-06	Leave Submission & Routing	Submit triggers auto-route to assigned reporting manager.	P0
FR-07	Attachment Upload	Optional PDF/JPG/PNG, max 5 MB, max 3 files. Stored on Uploadthing/Cloudinary.	P1
FR-08	Email Notification to Manager	Automated email via Resend with request details and secure Accept/Reject links. Links redirect to portal login → approval screen.	P1
FR-09	Manager Dashboard	Pending, approved, rejected tabs. Search/filter. Request detail view.	P0
FR-10	Manager Approve/Reject	One-click approve. Reject requires comment. Available on portal + email link.	P0
FR-11	2-Tier Routing (Manager → HR)	After manager approval, auto-route to HR. Leave confirmed only after both approve.	P0
FR-12	HR Dashboard	Org-wide view. Filters: department, status, date range, employee. Escalated items highlighted.	P0

FR-13	HR Final Approval/ Rejection	HR reviews manager-approved requests and gives final decision.	P0
FR-14	Employee Status Visibility	Real-time status tracking through all stages.	P0
FR-15	Employee Decision Notification	Portal + email notification on approval/rejection.	P1
FR-16	Leave History	Role-scoped: own (Employee), team (Manager), org (HR). Sortable, filterable.	P0
FR-17	Department Structure	8 departments: Finance, Marketing, IT, Procurement, Legal, Operations, Administration, Customer Support. Dynamic config.	P0
FR-18	In-Portal Comment Thread	Simple comment section per request. Employee ↔ Manager. Timestamped.	P1
FR-19	Audit Trail	Immutable log: actor, action, timestamp, previous status, new status.	P0
FR-20	Search, Filter, Sort	All views: search by name/ID, filter by status/dept/date/type, sortable columns.	P1
FR-21	Reports & Export	Department leave summary, CSV export, date range filter.	P1
FR-22	Auto-Reminder (48h)	Cron sends reminder if manager hasn't acted in 48 hours.	P1
FR-23	Auto-Escalation (72h)	Request flagged on HR dashboard if manager inaction > 72 hours.	P1

FR-24	Leave Cancellation	Employee can cancel before HR final approval. Logged in audit trail.	P0
FR-25	Responsive Design	Fully functional on desktop, tablet, and mobile browsers.	P0

11. Non-Functional Requirements

Req ID	Category	Requirement	Implementation
NFR-01	Security	Role-based access at API and UI level	Next.js middleware + API route guards
NFR-02	Security	Passwords hashed, sessions expire after 30 min idle	bcrypt hashing, JWT with 30-min expiry
NFR-03	Security	Email action links secured with signed tokens	JWT-signed URLs, 72-hour expiry, portal login required
NFR-04	Reliability	Leave requests and decisions must not be lost	PostgreSQL transactions with rollback
NFR-05	Auditability	All critical actions logged immutably	Separate audit_log table, append-only
NFR-06	Performance	Dashboard loads within 3 seconds	DB indexing, pagination, SSR
NFR-07	Scalability	Add departments/employees without code changes	Dynamic config, no hard-coded values

NFR-08	Usability	Submit leave in ≤ 3 clicks from dashboard	Clean UI with shadcn/ui
NFR-09	Usability	Responsive on all screen sizes	Tailwind CSS responsive utilities
NFR-10	Maintainability	Modular architecture for future HR modules	Separated modules, Prisma schema extensibility
NFR-11	Availability	≥ 99% uptime during business hours	Vercel edge network, managed DB
NFR-12	Data Privacy	Employee data accessible only by authorized roles	Field-level access, API-level role checks

12. Business Rules

Rule ID	Rule	Detail
BR-01	Portal is the single source of truth	All submissions via portal. Email = notification + quick-action only.
BR-02	2-tier approval mandatory	Every request needs Manager (Tier 1) + HR (Tier 2).
BR-03	Manager rejection is final	Rejected at Tier 1 → does not reach HR. Employee may resubmit.
BR-04	Rejection requires reason	Both manager and HR must provide a comment when rejecting.
BR-05	Leave balance enforced	Cannot submit if insufficient balance for that leave type.

BR-06	No past-dated requests	Start date must be today or future. HR override only for exceptions.
BR-07	No overlapping requests	Same employee cannot have two requests for the same date(s).
BR-08	Cancel before final approval only	Cancellation allowed only if HR has not yet approved.
BR-09	Auto-reminder at 48 hours	System emails manager if no action within 48h. (P1)
BR-10	Escalation at 72 hours	System flags request on HR dashboard if manager inaction > 72h. (P1)
BR-11	One department, one manager per employee	Every employee assigned to exactly one department and one reporting manager.
BR-12	Sick leave ≥ 3 days requires attachment	Medical certificate mandatory if SL exceeds 3 consecutive days. (P1)

13. Leave Type & Balance Model

Leave Type	Code	Annual Entitlement	Carry Forward	Encashable	Rules
Casual Leave	CL	12 days	No	No	Max 3 consecutive days per instance
Sick Leave	SL	12 days	No	No	Medical certificate if ≥ 3 consecutive days (P1)

Earned / Privilege Leave	EL	15 days	Yes (max 30 days)	Yes (on separation)	Apply ≥ 3 days in advance; accrued 1.25 days/month
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Balance Calculation:

- Opening Balance = Annual Entitlement + Carry Forward (EL only)
- Available Balance = Opening Balance - (Approved Leaves + Pending Leaves)
- Balance displayed before submission. Deducted only on **final HR approval**.

14. Approval Workflow — 2-Tier Model

Why 2-Tier

1. **Operational control** — Manager validates team/workload; HR validates policy/compliance.
2. **HR visibility** — Client's primary requirement; 2-tier makes HR an active participant.
3. **Audit strength** — Two independent approval actors = stronger trail.
4. **Industry standard** — Standard for Indian organizations with 100+ employees.

Workflow

Step	Actor	Action	Status Change	System Action
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1	Employee	Submits request	DRAFT → PENDING_MANAGER	Email to Manager (P1)
2	Manager	Reviews	—	Dashboard or email
3a	Manager	Approves	PENDING_MANAGER → PENDING_HR	Email to HR (P1)
3b	Manager	Rejects (with comment)	PENDING_MANAGER → REJECTED_BY_MANAGER	Email to Employee (P1). Flow ends.
4	HR	Reviews	—	HR Dashboard
5a	HR	Approves	PENDING_HR → APPROVED	Email to Employee (P1). Balance deducted.
5b	HR	Rejects (with comment)	PENDING_HR → REJECTED_BY_HR	Email to Employee + Manager (P1).
6	Employee	Cancels	→ CANCELLED	Audit trail updated.

Escalation (P1)

Condition	Action
Manager no action within 48h	Auto-reminder email
Manager no action within 72h	ESCALATED flag on HR Dashboard
HR no action within 48h	Auto-reminder email

15. Leave Status Model

Status	Code	Meaning	Next States
Draft	DRAFT	Being prepared	PENDING_MANAGER, CANCELLED
Pending Manager	PENDING_MANAGER	Awaiting manager	PENDING_HR, REJECTED_BY_MANAGER, ESCALATED, CANCELLED
Escalated	ESCALATED	Manager inaction > 72h	PENDING_HR, REJECTED_BY_MANAGER
Pending HR	PENDING_HR	Manager approved, awaiting HR	APPROVED, REJECTED_BY_HR, CANCELLED
Approved	APPROVED	HR approved, leave confirmed	Terminal
Rejected by Manager	REJECTED_BY_MANAGER	Manager rejected	Terminal
Rejected by HR	REJECTED_BY_HR	HR rejected	Terminal
Cancelled	CANCELLED	Employee cancelled	Terminal

16. Product Modules

Module	Capabilities	Priority	Sprint
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Authentication & Access	Login, logout, RBAC, JWT sessions	P0	Sprint 1
Employee Management	Profile, department assignment, manager linkage	P0	Sprint 1
Leave Management	Create, submit, view, track, cancel, balance display	P0	Sprint 2
Approval Engine	2-tier routing, approve/reject, comments, escalation (P1)	P0	Sprint 2
Manager Dashboard	Pending/approved/rejected, request details, actions	P0	Sprint 2
HR Dashboard	Org-wide, dept/status/date filters, escalated items	P0	Sprint 3
Notification Engine	Emails via Resend — submission, decisions, reminders	P1	Sprint 3
Comment Thread	Per-request comments, timestamped, append-only	P1	Sprint 3
Attachment Handler	Upload via Uploadthing/Cloudinary, URL in DB	P1	Sprint 3
Audit & History	Immutable audit_log, status transitions, leave history	P0	All sprints
Reports & Export	Dept summary, CSV export, date filter	P1	Sprint 3

17. Technology Stack

Layer	Technology	Rationale
Framework	Next.js 14+ (App Router)	Full-stack. Single codebase. SSR. Industry standard.

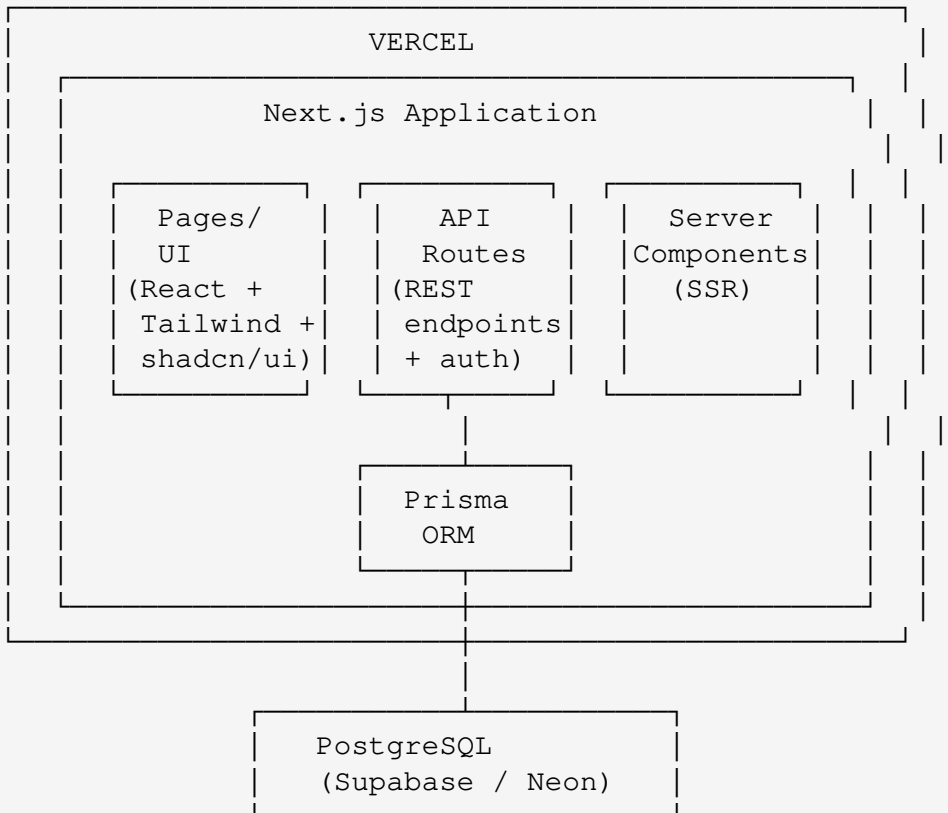
Language	TypeScript	End-to-end type safety.
Styling	Tailwind CSS	Rapid, responsive UI.
UI Components	shadcn/ui	Modern, accessible, professional look.
Database	PostgreSQL (Supabase or Neon)	Relational. Managed. Free tier.
ORM	Prisma	Type-safe queries. Auto migrations.
Authentication	NextAuth.js (Auth.js) v5	Credential login + role-based sessions via JWT.
Email	Resend	Free: 100 emails/day. React email templates.
File Storage	Uploadthing or Cloudinary	Free tier. Secure uploads.
Hosting	Vercel (free tier)	Optimized for Next.js. CDN. Auto-deploy. Free SSL.
DB Hosting	Supabase or Neon (free tier)	Managed PostgreSQL.
Cron Jobs	Vercel Cron	For 48h/72h reminders/escalation (P1).
Version Control	GitHub	<code>main</code> → <code>develop</code> → feature branches.
Agile Board	Jira	Industry-standard agile project management. Kanban + Scrum boards, sprint planning, backlog management, burndown charts.

Why This Stack

- **Single codebase** — No separate frontend/backend repos. Faster development.

- **₹0 hosting** — Every service has a free tier sufficient for this project's scale.
 - **Professional output** — shadcn/ui + Tailwind produces a clean, modern UI that doesn't look like a college project.
 - **Extensive documentation** — Every tech has massive community support and tutorials.
 - **Production-grade** — Same stack used by real startups and companies.
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18. System Architecture



```
Resend (Email)
- Notifications
- Action links
- Reminders
```

```
Uploadthing /
Cloudinary (Files)
```

Email Action Link Flow

```
Manager receives email → Clicks "Approve" link
→ Link: https://elap.vercel.app/api/action?token=<JWT>&ac
→ Portal verifies token (valid? not expired? correct mana
→ Redirects to portal login (if not logged in)
→ Auto-navigates to approval screen → Manager confirms →
```

19. Database Schema (High-Level)

Entity Relationships

```
Department (1) — (many) Employee
Employee (1) — (many) LeaveRequest
Employee (1) — (many) Employee (as reporting_manager)
LeaveRequest (1) — (many) Approval
LeaveRequest (1) — (many) Comment
LeaveRequest (1) — (many) Attachment
LeaveRequest (1) — (many) AuditLog
```

Key Tables

Table	Key Fields
department	id, name, created_at
employee	id, employee_id, name, email, password_hash, role (EMPLOYEE/MANAGER/HR), department_id, reporting_manager_id, joining_date
leave_type	id, code (CL/SL/EL), name, annual_entitlement, carry_forward, max_consecutive
leave_balance	id, employee_id, leave_type_id, year, opening_balance, used, pending, available
leave_request	id, employee_id, leave_type_id, start_date, end_date, days_count, reason, status, created_at, updated_at
approval	id, leave_request_id, approver_id, tier (MANAGER/HR), action (APPROVE/REJECT), comment, acted_at
comment	id, leave_request_id, author_id, content, created_at
attachment	id, leave_request_id, file_name, file_url, file_type, file_size, uploaded_at
audit_log	id, leave_request_id, actor_id, action, previous_status, new_status, timestamp

20. Constraints

#	Constraint	Impact
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C-01	Budget: ₹1,00,000 hypothetical (actual ≈ ₹0–3,000)	Free-tier stack eliminates cost concerns
C-02	Timeline: 5 weeks only (Aug 29 – Sep 30, 2026)	Aggressive. P0 is hard commitment. P1 = best effort.
C-03	Team: 5 members, limited active implementers	Realistic capacity planning critical
C-04	Academic context: ITM Skills University course project	Sponsor = Professor; constraints partly simulated
C-05	Web-only: Browser-based, no native app	Responsive web design compensates
C-06	No existing infrastructure: No servers, AD, or SSO	Standalone cloud-hosted



21. Assumptions



#	Assumption	Risk if Invalid
A-01	All users have modern browser + internet	Portal inaccessible
A-02	Resend free tier (100 emails/day) sufficient	Notifications blocked beyond limit
A-03	Standalone email/password auth sufficient (no SSO)	Additional auth work needed
A-04	8 departments: Finance, Marketing, IT, Procurement, Legal, Operations, Administration, Customer Support	Names may differ; system supports renaming
A-05	Leave: CL-12, SL-12, EL-15 (standard Indian policy)	Sponsor may specify different values

A-06	No integration with payroll/ERP/HRMS	API design needed if required
A-07	1:1 employee-to-manager mapping	Matrix reporting needs redesign
A-08	Vercel + Supabase/Neon free tiers sufficient	Paid tier if traffic exceeds limits
A-09	Prof. Mr. Sayantan available for UAT sign-off in final week	Gaps if unavailable
A-10	GitHub accessible to all team members	Alternate hosting needed

22. Dependencies

#	Dependency	Owner	Impact if Delayed
D-01	Professor confirms department names (or accepts assumptions)	Soham (BA) → Prof. Sayantan	Blocks dept seeding in Sprint 1
D-02	Resend account setup	Atharva (TL)	Blocks email module
D-03	Supabase/Neon account setup	Atharva (TL)	Blocks DB in Sprint 1
D-04	GitHub repo creation	Atharva (TL)	Blocks all development
D-05	Vercel project setup	Atharva (TL)	Blocks deployment
D-06	Professor confirms leave entitlements (or accepts standard)	Soham (BA) → Prof. Sayantan	Blocks balance model

23. Risk Register

ID	Risk	Prob.	Impact	Mitigation
R-01	Scope creep	High	High	Strict scope. PM (Sameer) gates all changes. Log for Phase 2.
R-02	2-tier workflow edge cases	Medium	High	Escalation rules defined. Edge cases in business rules.
R-03	Email action links blocked by spam	Medium	Medium	Portal is primary; email is convenience. All actions on portal.
R-04	Email link spoofing	Medium	High	JWT-signed, 72h expiry, portal login required.
R-05	Team capacity — 5-week timeline	High	High	P0 only is the hard commitment. P1 = best effort. Cut early.
R-06	Free-tier limits exceeded	Low	Medium	Scale is small. Monitor usage.
R-07	Fixed deadline — Sep 30, 2026	High	High	1-week sprints. Weekly reviews. Cut P1 at Sprint 2 review if behind.
R-08	Professor unavailable for clarification	Medium	Medium	Batch questions. Document assumptions.
R-09	Team member unavailability (exams, personal)	High	Medium	Front-load critical work in Sprint 1–2. Buffer in Sprint 4.

R-10	DB schema changes mid-development	Medium	Low	Prisma migrations handle evolution cleanly.
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24. Competitive Benchmarking

Tool	How ELAP Differentiates
SAP SuccessFactors	Too complex. ELAP = simpler, focused, zero learning curve.
PeopleForce	Mid-market. ELAP matches core features with cleaner, single-purpose design.
PeopleHR	Closest. ELAP adds in-portal discussion + email quick-actions.

ELAP's USP

- 1. **Single-purpose clarity** — One thing done extremely well.
 - 2. **In-portal discussion threads** — Communication inside the request.
 - 3. **Email quick-actions** — Accept/Reject from email, secured with JWT.
 - 4. **Zero training** — Submit in ≤ 3 clicks.
 - 5. **2-tier audit trail** — Complete traceability.
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25. Project Timeline & Sprint Plan

[!!IMPORTANT]

Total Duration: 33 days (Aug 29 – Sep 30, 2026)
Sprint Model: 1-week sprints
Decision Rule: If behind at any Sprint Review, P1 features are cut immediately.

Sprint	Dates	Goal	Deliverables	Owners
Sprint 0	Aug 29 – Sep 1 (4 days)	Discovery + Setup	BRD ✓, GitHub repo, Vercel + DB setup, Prisma schema, Jira board setup, project charter	Soham (BA), Sameer (PM), Antriksh (SA), Atharva (TL), Sanchita (SM)
Sprint 1	Sep 2 – Sep 8 (7 days)	Auth + Data Foundation	Login/logout, RBAC (Employee/Manager/HR), Employee + Department tables, seed 8 depts, employee profile page, role-based navigation	Atharva (TL), Antriksh (SA)
Sprint 2	Sep 9 – Sep 15 (7 days)	Leave Application + Approval Workflow	Leave form (type/dates/reason), balance display + enforcement, submission + routing, Manager dashboard, approve/reject, 2-tier routing to HR, HR approval screen, cancellation	Atharva (TL), Soham (BA)

Sprint 3	Sep 16 – Sep 22 (7 days)	HR Dashboard + P1 Features	HR org-wide dashboard (dept/ status/date filters), leave history, email notifications (Resend), attachment upload, comment thread, search/filter, CSV export, auto-reminder/escalation	Atharva (TL), Antriksh (SA)
Sprint 4	Sep 23 – Sep 30 (8 days)	Testing + UAT + Polish + Go-Live	End-to-end testing, responsive testing, bug fixes, UI polish, UAT with Prof. Sayantan, final deployment, project closure report	All team

Sprint Ceremonies

Ceremony	When	Duration	Format
Daily Standup	Daily	5 min	WhatsApp text: Did / Doing / Blocked
Sprint Planning	Monday of each sprint	20 min	Google Meet or in-person
Sprint Review	Last day of sprint	20 min	Demo to team (+ professor if available)
Retrospective	Last day of sprint	10 min	Quick WhatsApp/Notion reflection

Estimation: T-Shirt Sizes

Size	Effort	Example
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S	< 2 hours	Add a filter dropdown
M	2–4 hours	Build leave request form
L	4–8 hours	Build manager dashboard
XL	8+ hours	Implement 2-tier approval engine

Definition of Done

A feature is "Done" when:

- ☐ Functional requirement met per BRD
 - ☐ Acceptance criteria passed
 - ☐ UI is responsive (desktop + mobile)
 - ☐ RBAC verified (no unauthorized access)
 - ☐ Audit trail logs correctly
 - ☐ No critical bugs
 - ☐ Deployed to Vercel
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26. Budget Estimate

Category	Cost (₹)	Notes
Vercel Hosting	0	Free tier
Supabase / Neon	0	Free tier
Resend	0	Free: 100 emails/day
Uploadthing / Cloudinary	0	Free tier
GitHub	0	Free
Domain (optional)	500–1,000	Vercel provides <code>.vercel.app</code> URL
Miscellaneous	500–2,000	Documentation, minor tools

Total Actual	₹500 – ₹3,000	
Hypothetical Budget	₹1,00,000	
Utilization	< 3%	Cost-efficiency demonstrated

27. Testing Strategy

Type	Approach	Owner	When
Functional Testing	Manual testing against acceptance criteria	Soham (BA)	Sprint 2–4
RBAC Testing	Verify each role's access boundaries	Soham (BA)	Sprint 1, 2, 3
Workflow Testing	End-to-end: submit → manager approve → HR approve	Soham (BA)	Sprint 2–3
Responsive Testing	Desktop, tablet (iPad), mobile (iPhone, Android Chrome)	Soham (BA)	Every sprint
Email Testing	Verify Resend sends correctly	Atharva (TL)	Sprint 3
Edge Case Testing	Overlapping dates, insufficient balance, cancellation timing, escalation	Soham (BA)	Sprint 3–4
UAT	Prof. Sayantan tests key scenarios on staging	Soham (BA) + Prof. Sayantan	Sprint 4

Automated testing (unit/integration) is optional — added only if time permits. Primary strategy is **manual testing by BA**.

28. Acceptance Criteria & UAT

28.1 Go/No-Go Criteria

#	Criterion	Verification
AC-01	Employee can log in and submit a leave request	UAT
AC-02	Leave balance displayed and enforced correctly	Data verification
AC-03	Request routes to correct reporting manager	Multi-dept UAT
AC-04	Manager can approve/reject with comments	UAT
AC-05	Approved request routes to HR for final approval	UAT
AC-06	HR can approve/reject; employee is notified	UAT
AC-07	HR dashboard shows all 8 departments with filters	Dashboard demo
AC-08	Balance deducted only on final HR approval	Data verification
AC-09	Audit trail exists for every status change	Audit log review
AC-10	Portal is responsive on mobile	Device testing
AC-11	No Excel/email needed for leave management	Stakeholder confirmation

28.2 UAT Process

Step	Activity	Owner
1	Prepare UAT test scripts	Soham Ahirrao (BA)
2	Schedule UAT with professor	Sameer Rathod (PM)
3	Professor tests on staging	Prof. Sayantan + Soham
4	Defects logged and fixed	Atharva Gahine (TL)
5	Re-test	Soham (BA)
6	Formal sign-off	Prof. Mr. Sayantan

29. Phase 2 Roadmap (Future Scope)

#	Feature	Complexity	Value
1	Attendance & Time Tracking	High	Complements leave management
2	Leave Balance Auto-Accrual (monthly)	Medium	Eliminates manual balance setup
3	Payroll Integration	High	Leave → salary processing
4	Native Mobile App (React Native)	High	Better mobile experience
5	SSO / Active Directory	Medium	Enterprise auth
6	Advanced Analytics	Medium	Data-driven workforce planning
7	Performance Management	High	Broader HRMS
8	Recruitment & Onboarding	High	Full HRMS suite
9	Multi-language Support	Low	Accessibility

10	Manager Delegation (alternate approver)	Medium	Better unavailability handling
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30. Glossary

Term	Definition
ELAP	Employee Leave Management & Approval Portal
BRD	Business Requirements Document
CL	Casual Leave — 12 days/year
SL	Sick Leave — 12 days/year
EL	Earned/Privilege Leave — 15 days/year
RBAC	Role-Based Access Control
2-Tier	Manager + HR sequential approval
MVP	Minimum Viable Product (P0 features)
MoSCoW	Must/Should/Could/Won't prioritization
UAT	User Acceptance Testing
P0	Must Have for launch
P1	Important; best-effort within timeline
P2	Future scope — excluded from Phase 1
JWT	JSON Web Token
ORM	Object-Relational Mapping (Prisma)
SSR	Server-Side Rendering (Next.js)
CDN	Content Delivery Network (Vercel)
SMTP	Simple Mail Transfer Protocol
SSO	Single Sign-On
API	Application Programming Interface

DoD	Definition of Done
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31. Sign-Off & Approval

By signing below, the stakeholder confirms that the business requirements in this BRD are accurate, complete, and approved for the team to proceed.

Role	Name	Signature	Date
Project Sponsor	Prof. Mr. Sayantan		
Project Manager	Sameer Rathod		
Business Analyst	Soham Ahirrao		
Solution Architect	Antriksh Manwadkar		
Tech Lead	Atharva Gahine		
Scrum Master	Sanchita Suryawanshi		

Document End — BRD-ELAP-001 V1.0

Prepared by: Soham Ahirrao, Business Analyst Date: August 29, 2026